

Covalent bonding

Learning Objective:

Know that covalent bonds are based on sharing electrons

Draw a diagram to show the formation of covalent bonds

Know that covalent compounds can exist in small molecules or giant covalent structure



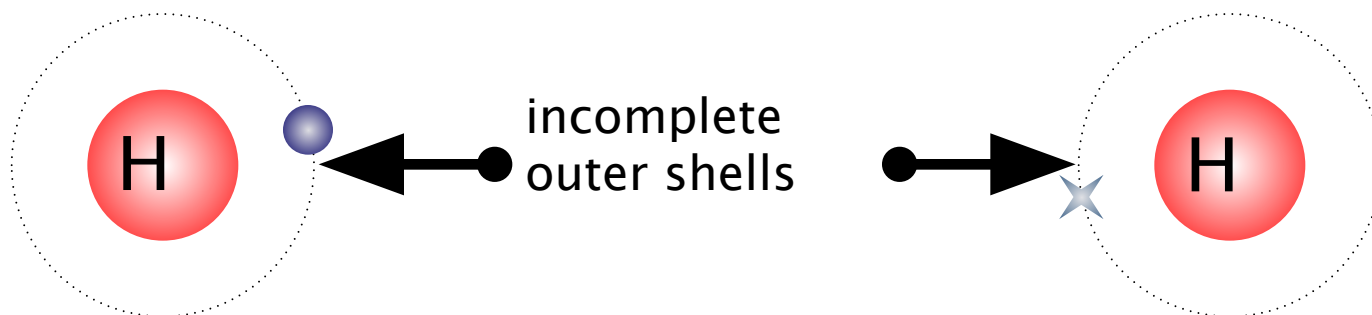
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Do Now

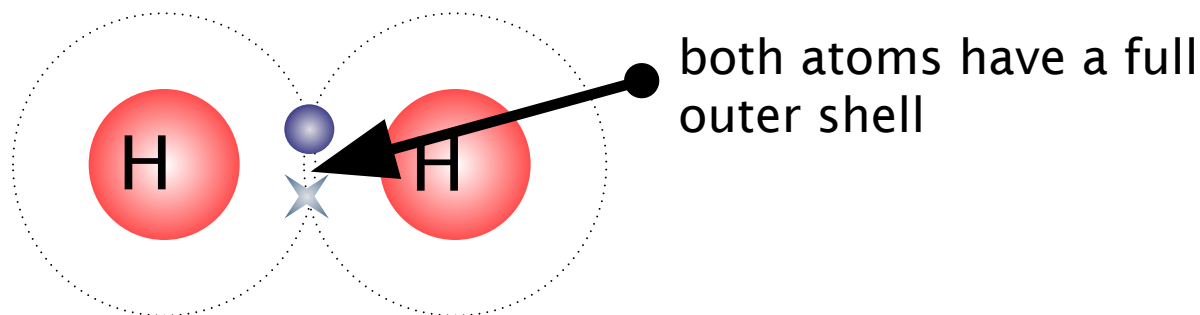
- Revision. How many facts can you recall about Ionic Bonding?
1. Ionic bonding occurs between a metal and non-metal.
 2. Metals give away electrons and a non-metals accept electrons leaving both ions with stable full shells.
 3. Ions formed by losing electrons have positive charge, where ions formed by gaining electrons have negative charge

The covalent bond

- When non-metal atoms react together, they need to gain electrons to fill their outer shell and become stable.



They can only do this if they share electrons with each other.



The atoms share electrons so there is a strong force that joins the atoms together. This is called a **covalent bond**.

Covalent bonding

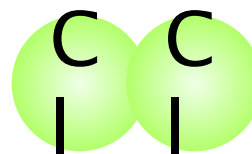
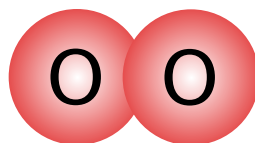
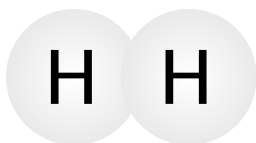
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graph TD; A[Covalent bonding] --> B[Hold two or three atoms together forming simple molecule (hydrogen, chlorine, water, methane, ammonia)]; A --> C[Form compound (giant structures) made of many atoms together (diamond, polymers, silicon dioxide)];
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Hold two or three atoms together forming **simple molecule** (hydrogen, chlorine, water, methane, ammonia)

Form compound (**giant structures**) made of many atoms together (diamond, polymers, silicon dioxide)

Covalent bonding and elements

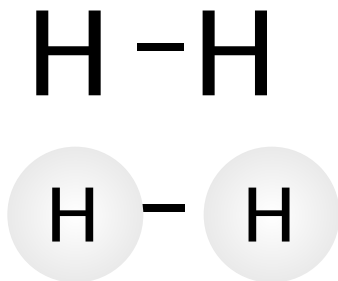
Many elements exist as **molecules** – two or more atoms joined by a covalent bond. Each atom has a full outer electron shell and is therefore stable.



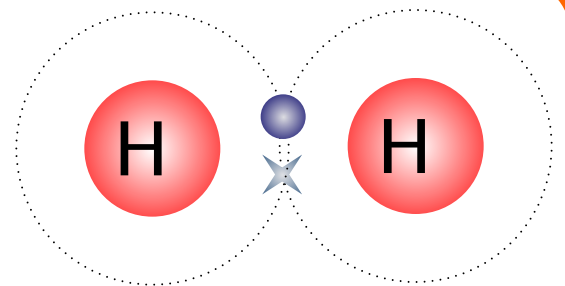
Only the outer shell of electrons is involved in covalent bonding. This means that the inner shells do not always have to be included in diagrams.

Two common ways to indicate a covalent bond are:

solid line



dot and cross
diagram

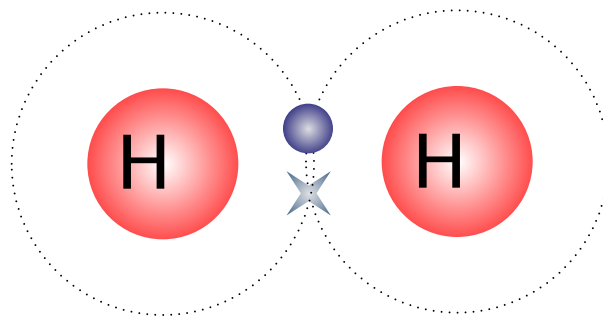


Covalent bonding in hydrogen



Hydrogen (electron configuration: 1) needs 1 more electron to have a completely full outer shell.

To achieve this, it can share an electron with another hydrogen atom. This creates a **single bond** and the two hydrogen atoms form a hydrogen molecule.



There are two atoms in the molecule so it is called diatomic.

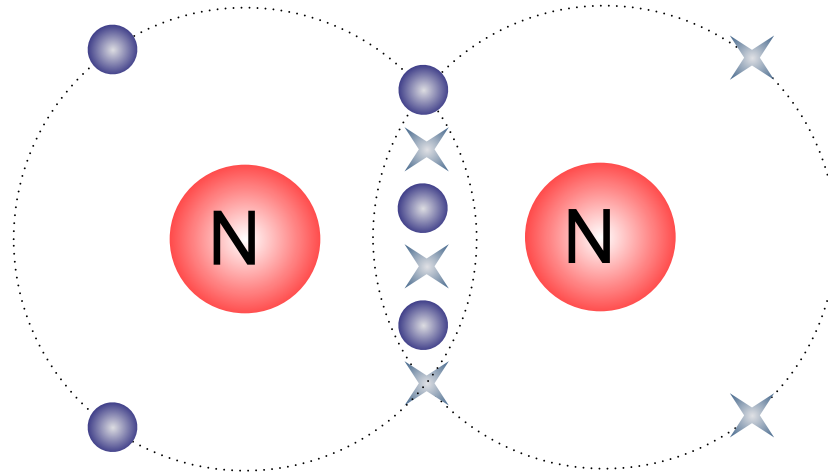
Covalent bonding in

nitrogen



Nitrogen (2.8.5) needs 3 more electrons to have a completely full outer shell.

It can share three electrons with another nitrogen atom to do this. This creates a **triple bond**.

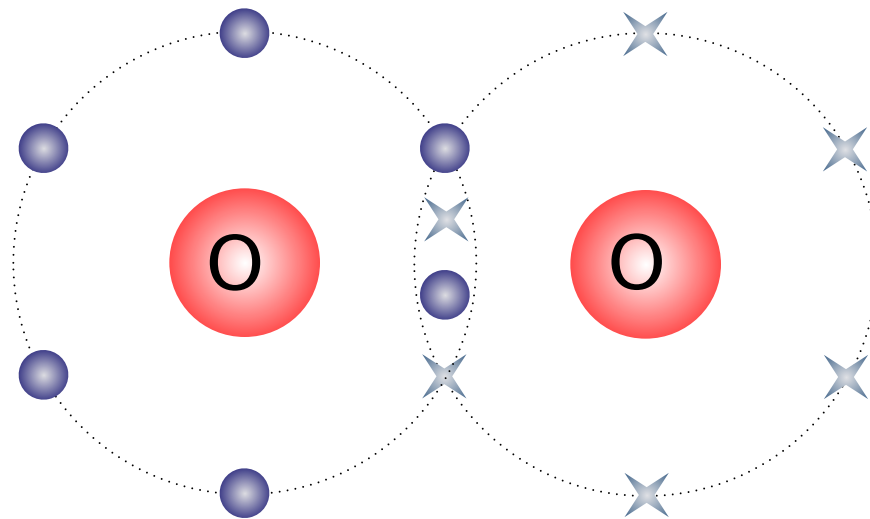


- Draw the covalent bond between two oxygen atoms

Covalent bonding in oxygen

- Oxygen (2.8.6) needs 2 more electrons to have a completely full outer shell.

To achieve this, it can share two electrons with another oxygen atom. This creates a **double bond**.



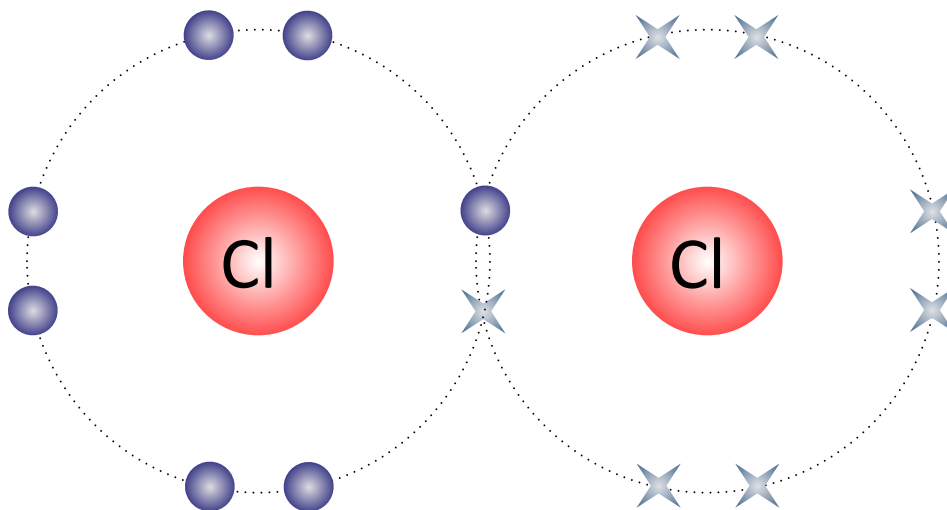
- Your turn:

- Draw the covalent bond between two chlorine atoms

Covalent bonding in chlorine

Chlorine (2.8.7) needs 1 more electron to have a completely full outer shell.

To achieve this, it can share an electron with another chlorine atom. This creates a single bond.



Covalent bonding in compounds. Further practice.

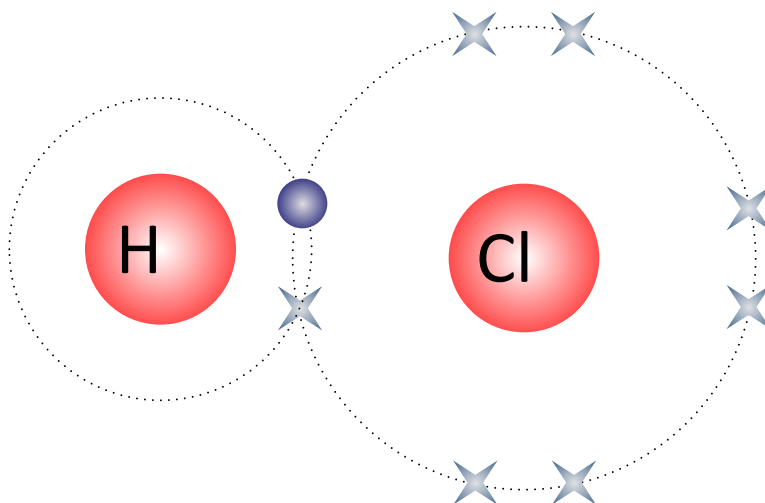
- Draw the covalent bonding in hydrochloric acid HCl, Water H₂O, Ammonia NH₃ and Methane CH₄.

Covalent bonding in compounds



Covalent bonding can take place between atoms of different elements to create molecules of **covalent compounds**. These covalent bonds can be **single, double or triple**.

Both hydrogen (1) and chlorine (2.8.7) need 1 more electron to fill their outer shell. By sharing one electron each, they can fill their outer shells and become stable.

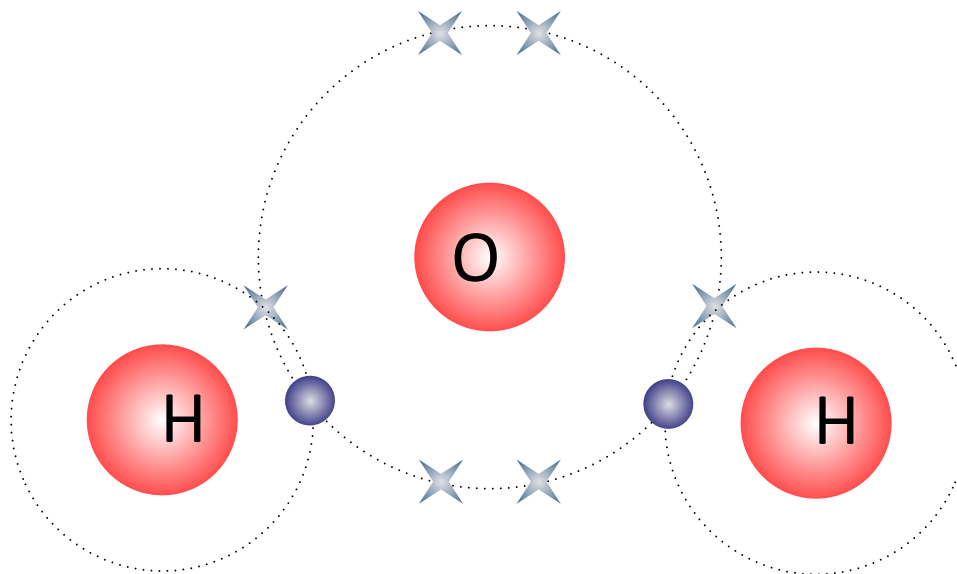


HCl or H-Cl

Covalent bonding in water

- Oxygen (2.6) needs 2 more electrons, but hydrogen (1) only needs 1 more. How can these two elements be covalently bonded?

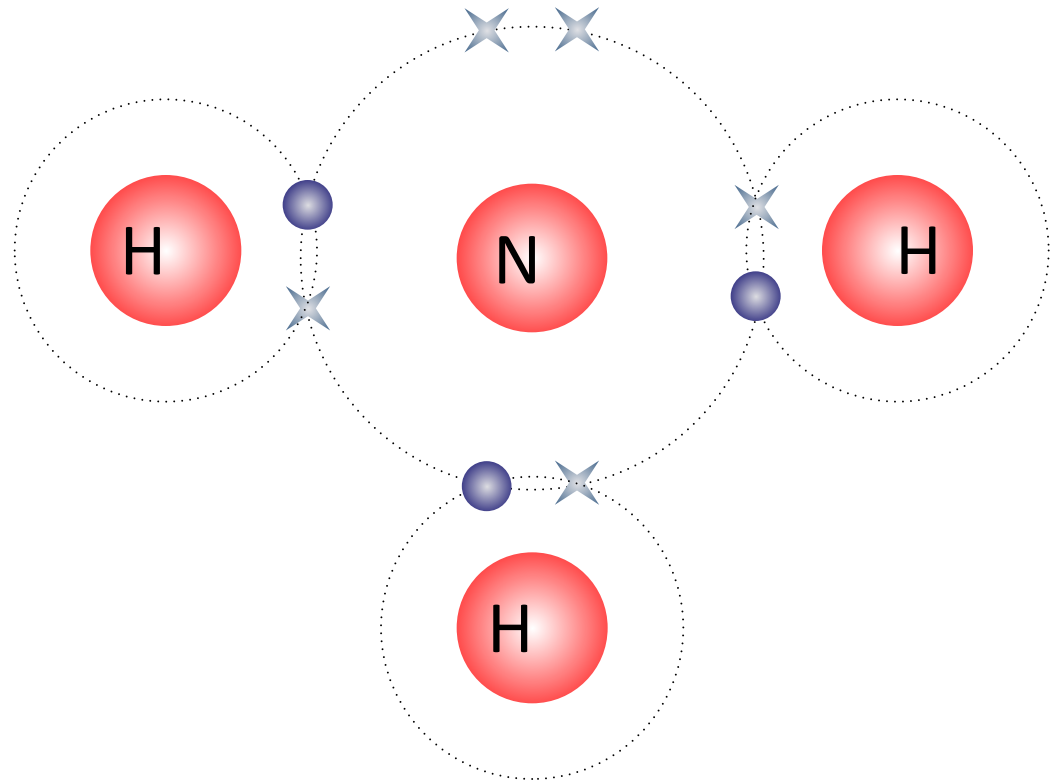
The oxygen atom shares 1 electron with 1 hydrogen atom, and a second electron with another hydrogen atom.



Covalent bonding in ammonia



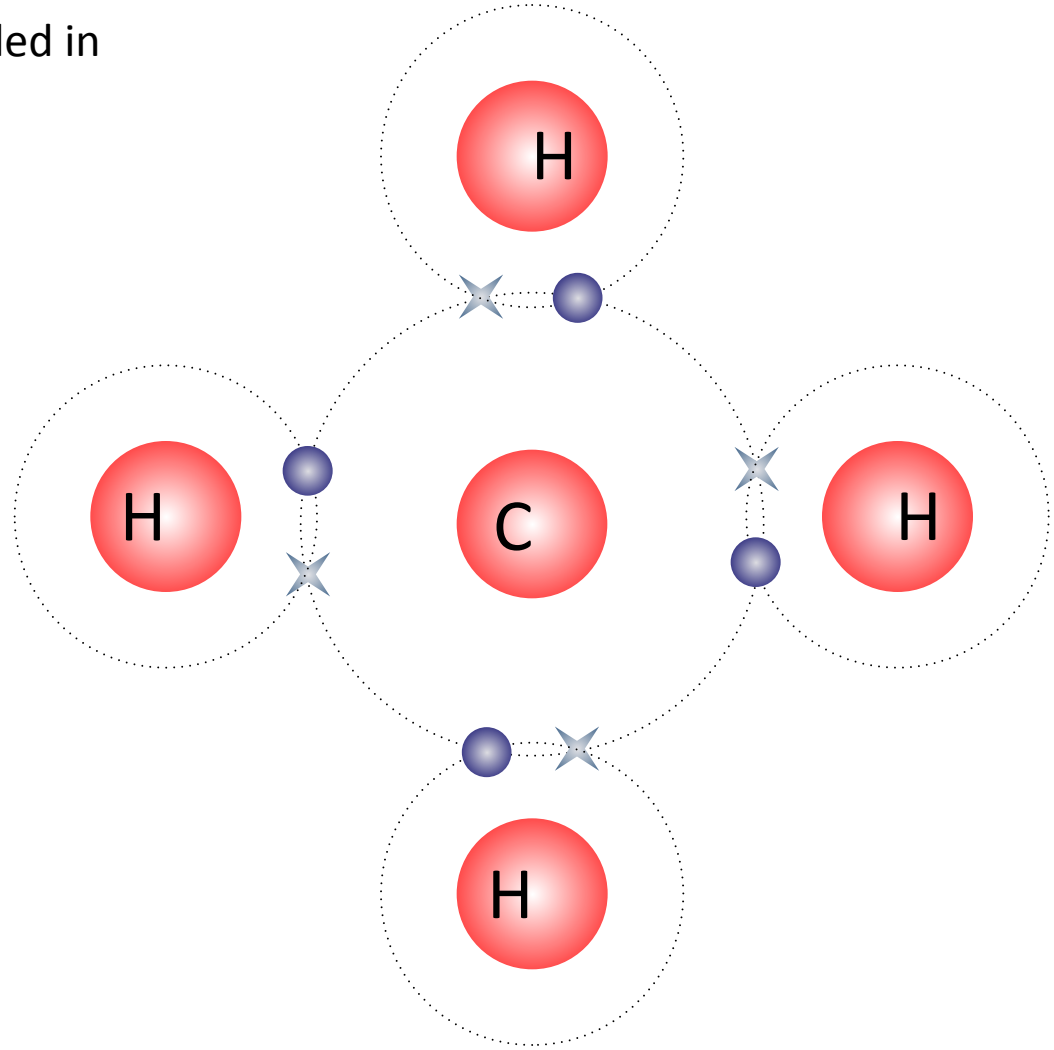
How are nitrogen and hydrogen bonded in ammonia NH_3 ?



Covalent bonding in methane



How are carbon and hydrogen bonded in methane?



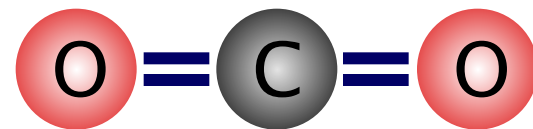
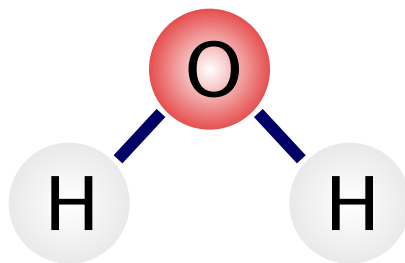
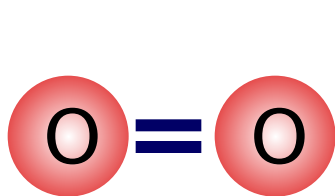
Title : Covalent bonding and giant structures

- LO
- Know that covalent compounds can exist in small molecules or giant covalent structure

Simple covalent structures

Atoms that join together by covalent bonding can form different types of covalent structure.

Oxygen, water and carbon dioxide are molecules. They have a simple structure because they only contain a few atoms.



Most molecular substances are gas or liquid at room temperature. A few are solid and these are called **molecular solids**.

Properties of simple covalent structures

The weak forces of attraction between molecules in molecular solids only require a small amount of energy to be broken. This means that molecular solids:

- have low melting and boiling points;
- are usually soft and brittle – they shatter when hit.

In addition, molecular solids:

- are usually insoluble in water but soluble in other solvents such as petrol;
- cannot conduct electricity – there are no free electrons to carry an electrical charge.

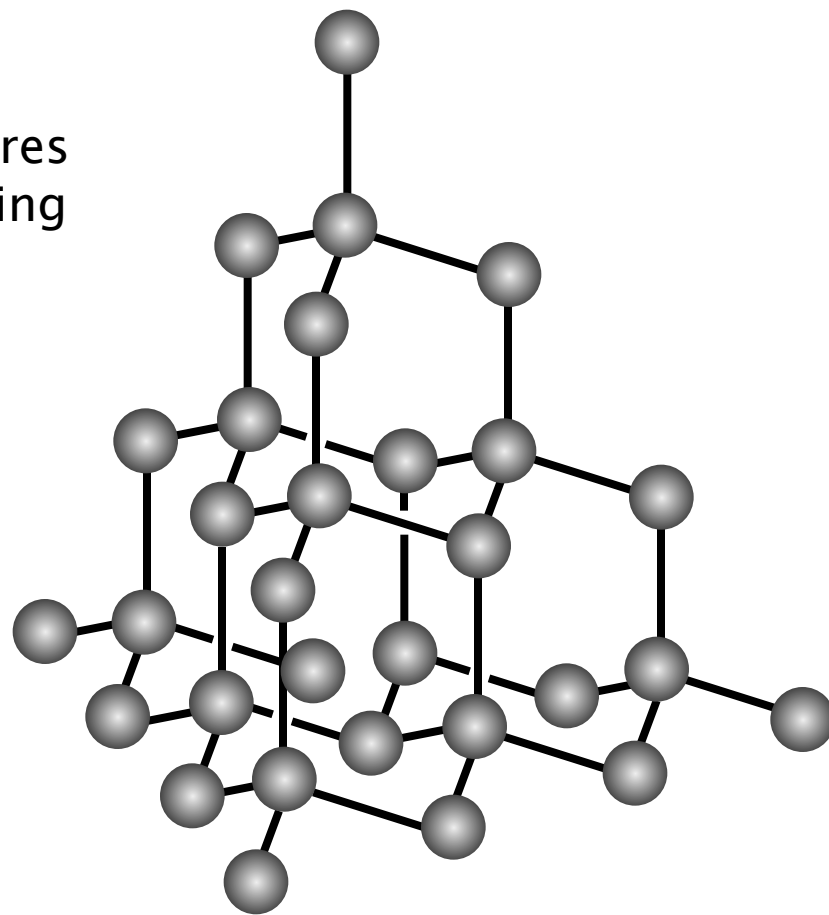
Giant covalent

structures

In some substances, millions of atoms join together by covalent bonding. This produces **giant covalent structures**, not molecules.

All the bonds are covalent, which means that giant covalent structures have a very high melting and boiling point.

This also means that almost all giant covalent structures are hard but brittle.



Allotropes of carbon



In the element carbon, atoms bond in different ways, creating different kinds of giant structures.

Two of these structures are diamond and graphite. They are called **allotropes** of carbon.

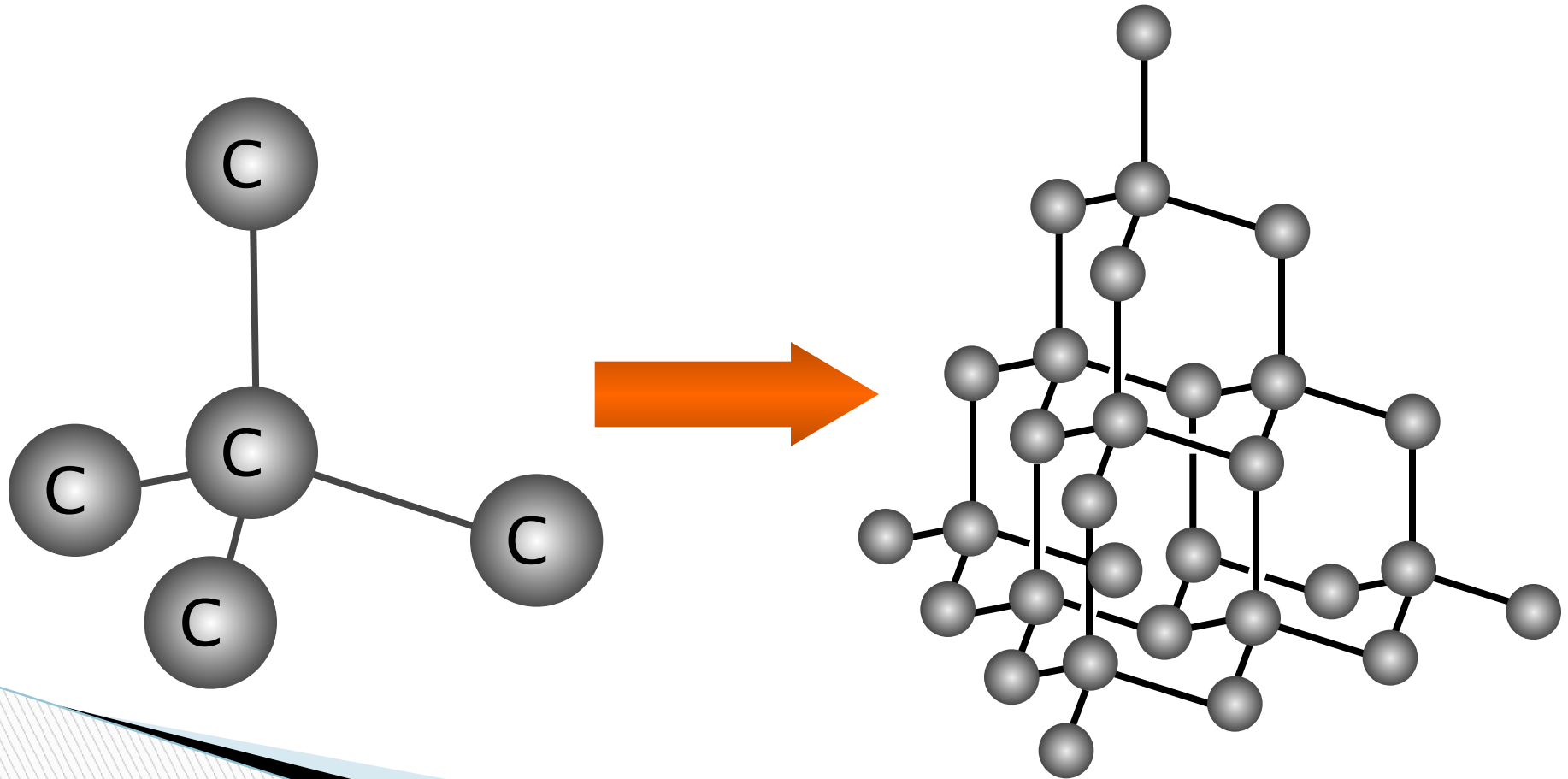
Allotropes have the same chemical properties because they have the same number of electrons.

However, they have different physical properties because the electrons are shared in different ways with other atoms.

The structure of diamond

Diamond is a rare form of carbon in which each atom is covalently bonded to four others.

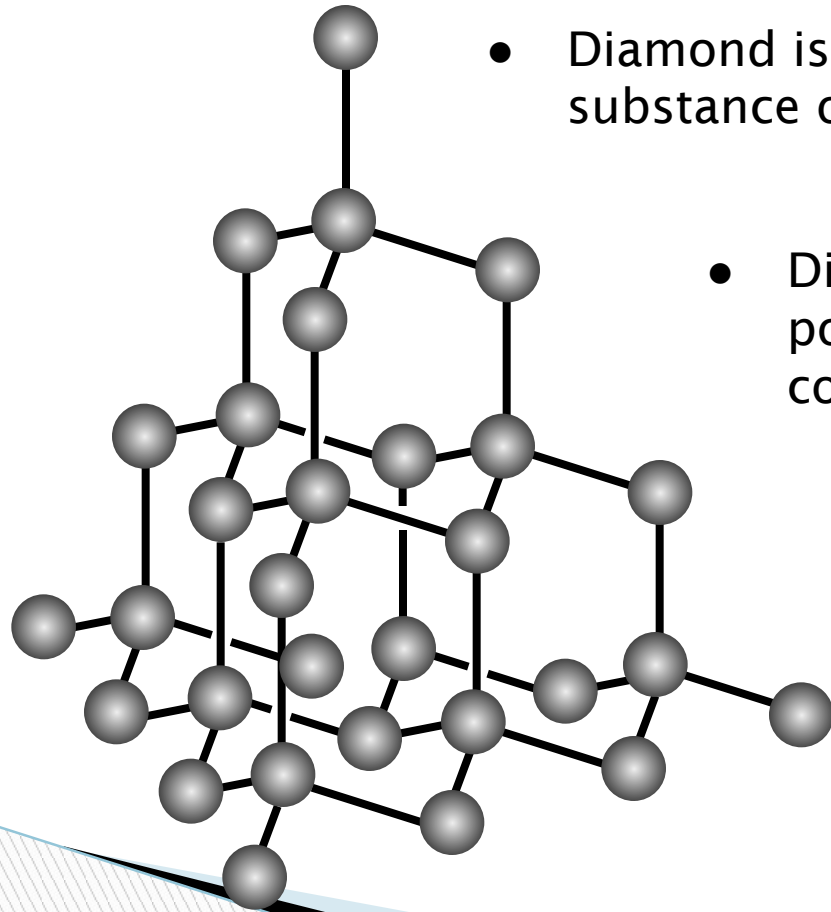
This pattern arrangement is repeated millions of times to create a giant lattice.



The properties of

diamond

All the electrons in the outer shell of the carbon atom (2.4) are used in covalent bonds. This affects diamond's properties.



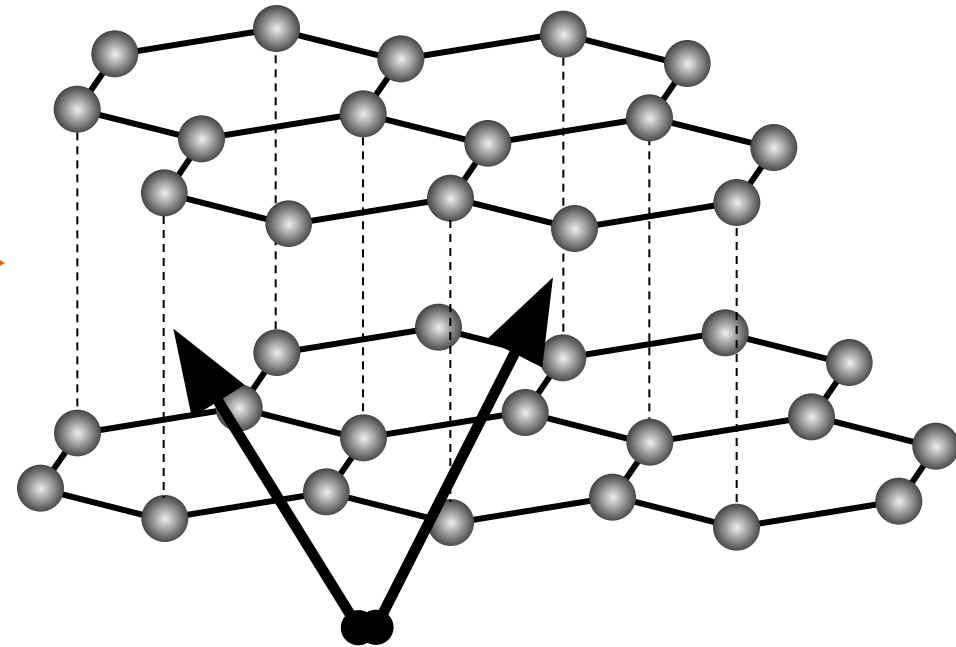
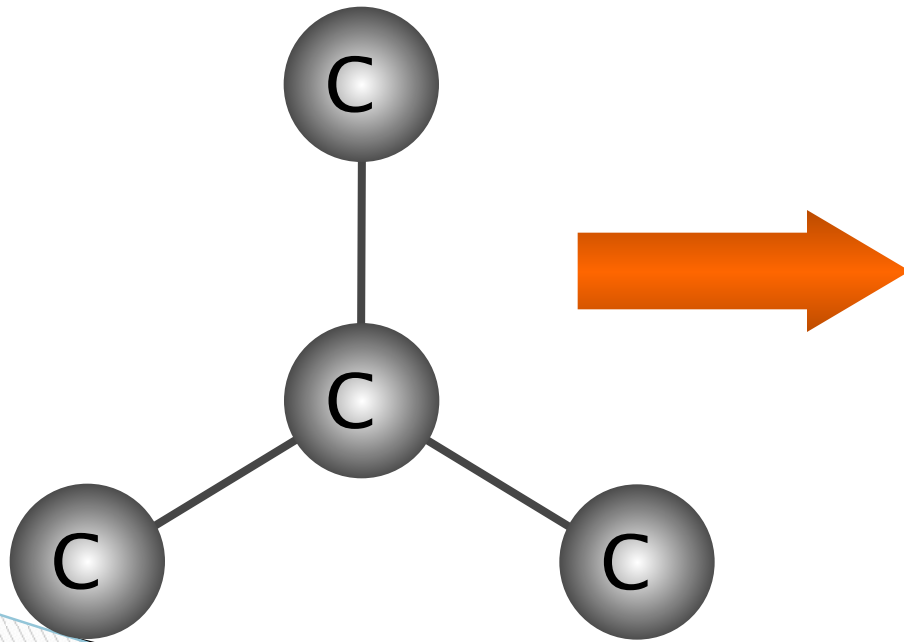
- Diamond is very hard – the hardest natural substance on Earth.
- Diamond has a very high melting and boiling point – a lot of energy is needed to break the covalent bonds.
- Diamond cannot conduct electricity – there are no free electrons or ions to carry a charge

The structure of

graphite

Graphite is a much more common form of carbon. in which each atom is covalently bonded to three others.

This forms rings of six atoms, creating a giant structure containing many layers. These layers are held together by weak forces of attraction.



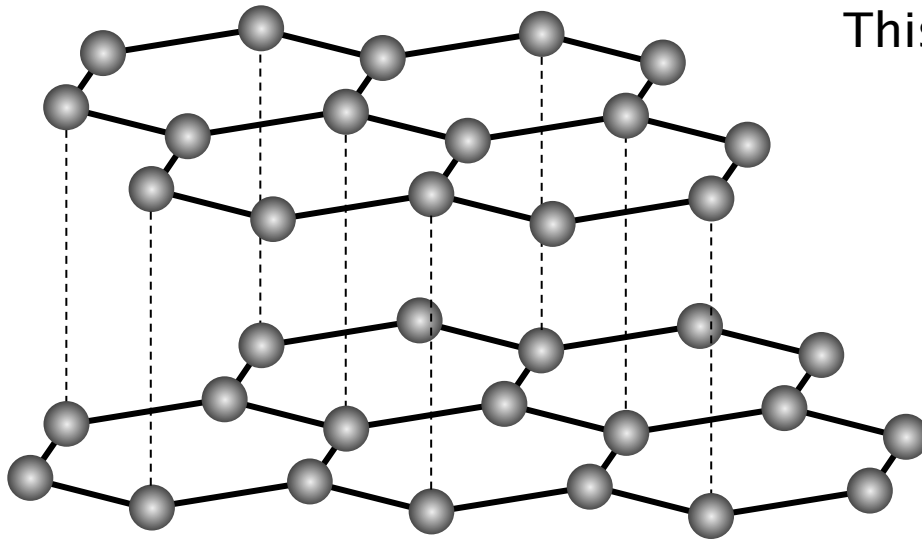
weak forces of attraction

The properties of

graphite

Only three of the four electrons in the outer shell of the carbon atom (2.4) are used in covalent bonds. This affects graphite's properties.

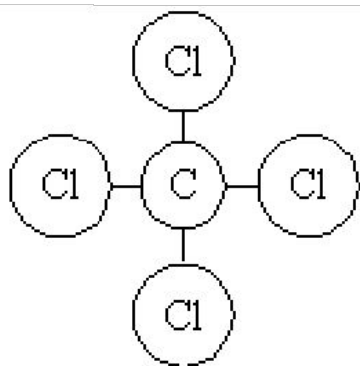
- Graphite is soft and slippery – layers can easily slide over each other because the weak forces of attraction are easily broken. This is why graphite is used as a lubricant.



- Graphite can conduct electricity – the only non-metal to do. There is a free electron from each atom to carry a charge.

Practice questions

1. Chlorine will combine with the non-metal element, carbon, to form this molecular compound:



- (a) What is the symbol equation for this molecule?

.....

..... (1)

- (b) Explain how these bonds are formed. (You may use a diagram).

.....

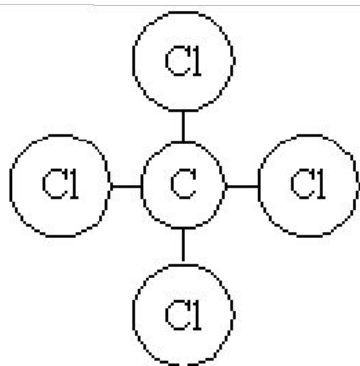
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..... (2)

(Total 3 marks)

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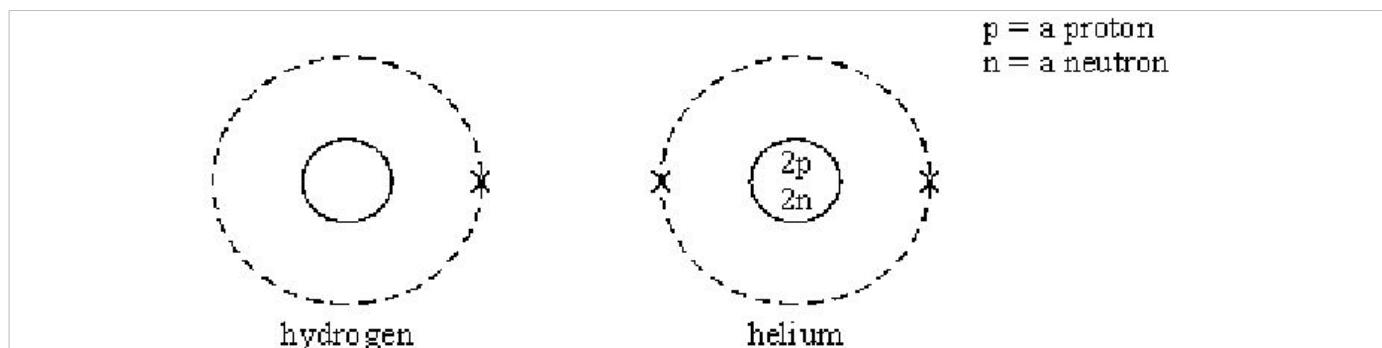
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- (b) Explain how these bonds are formed. (You may use a diagram).

.....
.....
.....
..... (2)

(Total 3 marks)

1. (a) The diagrams represent the atomic structures of two gases, hydrogen and helium.



Hydrogen gas is made up of diatomic molecules (molecules with two atoms).
Helium gas exists as single atoms.

- (i) How is a molecule of hydrogen formed from two hydrogen atoms?
(You may use a diagram as part of your answer)

.....

.....

.....

(2)

- (ii) Why does helium exist only as single atoms?

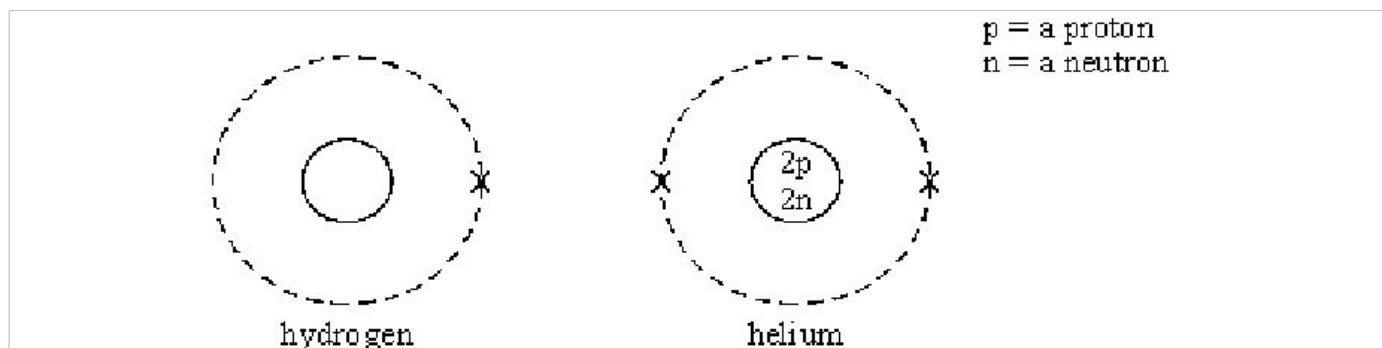
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(2)

The table gives information about the atoms of three elements.

Name of element	Chemical symbol	Number of electrons in:		
		1 st shell	2 nd shell	3 rd shell
Fluorine	F	2	7	0
Neon	Ne	2	8	0
Sodium	Na	2	8	1

Two of these elements can react together to form a chemical compound.

(a) What is the name and the formula of this compound?

Name

.....

Formula

.....

(2)

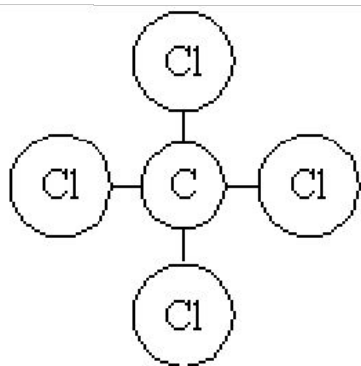
(b) What type of bonding holds this compound together?

.....

(1)

(Total 3 marks)

1. Chlorine will combine with the non-metal element, carbon, to form this molecular compound:



- (a) What is the symbol equation for this molecule?

CCl_4

(1)

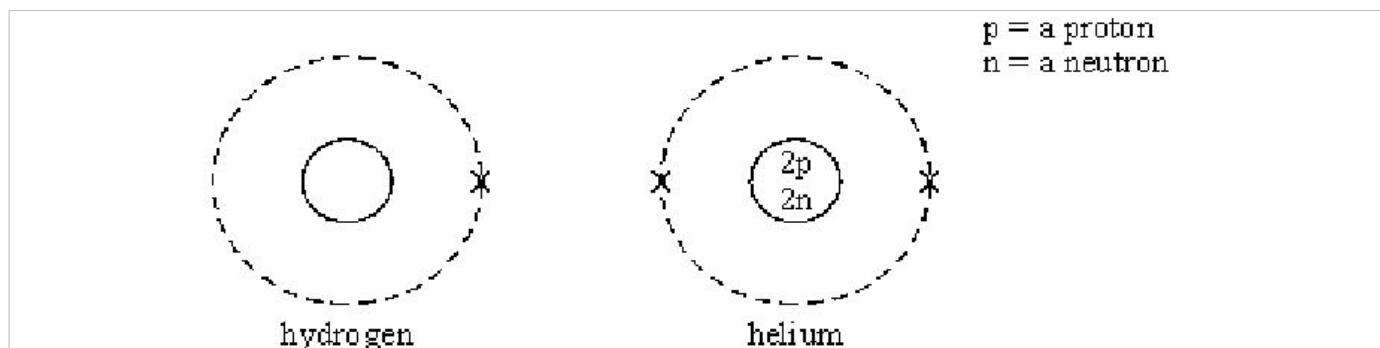
- (b) Explain how these bonds are formed. (You may use a diagram).

The bonds between carbon and chlorine atoms are covalent bonds. Bonds are formed by sharing one pair of electrons between carbon atom and each chlorine atom.

(2)

(Total 3 marks)

1. (a) The diagrams represent the atomic structures of two gases, hydrogen and helium.



Hydrogen gas is made up of diatomic molecules (molecules with two atoms). Helium gas exists as single atoms.

- (i) How is a molecule of hydrogen formed from two hydrogen atoms?
(You may use a diagram as part of your answer)

Hydrogen molecule is formed from two hydrogen atoms by sharing electrons. Hydrogen atom has single electron in the outer shell and is unstable. To gain stability each hydrogen atom share its single electron with another hydrogen.

(2)

- (ii) Why does helium exist only as single atoms?

Helium is a noble gas. It is atom has two electrons in the outer shell. The helium atom is stable as it has full of electrons outer shell. That why it can exist only as a single atom.

The table gives information about the atoms of three elements.

Name of element	Chemical symbol	Number of electrons in:		
		1 st shell	2 nd shell	3 rd shell
Fluorine	F	2	7	0
Neon	Ne	2	8	0
Sodium	Na	2	8	1

Two of these elements can react together to form a chemical compound.

(a) What is the name and the formula of this compound?

Name **Sodium fluoride**

.....

Formula **NaF**

.....

(2)

(b) What type of bonding holds this compound together?

Ionic bonds by transferring electrons

.....

(1)

(Total 3 marks)